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Problem Statement

- Manual expense settlement between friends/groups is tedious and error-prone.
- Users rely on WhatsApp or memory to track who owes how much to whom.
- Existing apps like Splitwise only show due amounts; actual payments are still manual.
- No proper real-time UPI-based auto-tracking or smart features for personalized splits.

Proposed Solution

- SplitKaro is a web-based FinTech platform for smart group expense management.
- Users can create groups, add expenses, and split them based on custom logic.
- Includes a 'Pay Now' UPI Deeplink feature for one-tap settlement via GPay/PhonePe.
- System automatically checks UPI transaction status and marks expense as settled.

Features

- Auto UPI Payment via Pay Now button with dynamic deeplink.
- Smart Split logic: unequal share, itemized splits, usage-based allocation.
- Gamified Trust Score: track who pays on time, badges for fun social engagement.
- Voice command support for adding expenses quickly.
- Monthly summary + PDF export and reminder notifications.

Tech Stack

- Frontend: React.js + Tailwind CSS / Shadcn UI
- Backend: Node.js with Express.js
- Database: MongoDB (Mongoose)
- Payment: UPI Deeplink, Razorpay / PhonePe APIs
- Others: Cron jobs, JWT Auth, Web Speech API (voice input), Webhooks

Workflow Diagram

- 1. User logs in & creates group.
- 2. Adds expenses (via form or voice input).
- 3. App calculates split and shows pending amounts.
- 4. Pay Now button opens UPI app with prefilled details.
- 5. Backend checks/ref ID to mark settled.
- 6. Dashboard updates & trust score adjusts.

Target Audience and Engagement

- Target: College students, flatmates, frequent travellers, office teams.
- Engagement: Fun badges, clean UI, minimal steps to pay.
- Social sharing of trust score or group stats.
- Privacy-focused: no ads, no unnecessary data tracking.
- Can go viral in student communities with referral rewards.

Future Scope

- AI-based smart splitting suggestions (based on past behavior).
- Razorpay payment link integration with auto reconciliation.
- Global multi-currency support for international groups.
- WhatsApp bot for adding/viewing expenses directly from chat.
- Mobile app with offline mode (PWA support).

AI Integration in SplitKaro

- AI-powered Smart Split: Automatically suggest split ratios based on past group behavior.
- Personalized Recommendations: Suggest optimal settlement paths to reduce total number of transactions.
- Voice Input Understanding: Use NLP to parse spoken expense inputs accurately.
- Fraud Detection: Detect unusual payment behaviors using anomaly detection algorithms.
- Expense Categorization: Auto-categorize expenses using trained ML models for better reporting.
- Trust Score AI: Adjust user trust scores dynamically using AI based on payment timeliness.